



The impact of school choice and public policy on segregation: Evidence from Chile

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ABSTRACT

Advocates argue that vouchers can make improved educational opportunity available to disadvantaged students. Critics contend that vouchers increase the risk of stratification. Researchers have found that Chile's voucher program has led to increased socioeconomic school segregation. What has been overlooked, however, is segregation between schools within a sector and variation within private for-profit and non-profit school sectors. I find that public schools are more likely to serve disadvantaged students than private voucher schools. I also find that disadvantaged students are more segregated among private voucher schools than among public schools. While between and within sector segregation levels vary across private voucher school types, the differences are not always consistent with theory. The data also suggest that policies can either mitigate or exacerbate the stratifying effects of educational vouchers.

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1. Introduction

School choice is perhaps the most hotly debated issue in current education policy reform discussions. Advocates have often argued that the introduction of vouchers can make improved educational opportunity available to the most disadvantaged children (Sugarman, 1999). Since the option of school choice through residential mobility or through enrollment in private schools has long been available to higher income parents, voucher proponents maintain that expanding the right of minorities and parents of lower socioeconomic status to leave their low performing local schools for higher performing ones may reduce school segregation (Neal, 2002; Nechyba, 2000).

Voucher skeptics contend that vouchers increase the risk of exacerbating inequities based on race and socioeconomic status (Fiske and Ladd, 2000; Levin, 1998; Henig, 1994). Critics have identified both demand and supply side reasons why educational vouchers may lead to more stratified schools. On the demand side, they are worried about whether minority and less educated families have enough information to make informed decisions (Schneider et al., 2000) and whether parental preferences will lead parents to select schools based on the race or class composition of their student bodies and not on their academic quality (Saporito, 2003). On the supply side, voucher critics argue that in order to remain competitive and save costs, schools have incentives to skim off the highest performing students who are usually least demanding in terms of resources (Epple and Romano, 1998).

Researchers can gain insight into this debate by examining school systems where vouchers have been implemented on a large scale and where private school supply has increased. In 1981, Chile began financing public and most private schools with vouchers. Private voucher schools currently account for over 43 percent of total enrollments.

Over the last three decades, the Chilean system has generated a significant cottage industry of scholarly research that has yielded a range of conflicting findings. While there is still debate on whether or not almost 30 years of vouchers has improved student achievement,¹ most evidence suggests that unrestricted choice in Chile has exacerbated stratification between public and private schools. Researchers have found that parents search for instructional environments in which their children attend schools with peers of similar social backgrounds (Elacqua et al., 2006) and that private voucher schools “cream skim” off middle and high-income students while the most disadvantaged students are more likely to attend public schools (Hsieh and Urquiola, 2006).

Voucher critics often hold up these results as proof that unfettered choice will increase stratification (OECD, 2004). However, the under-representation of disadvantaged students in

¹ For instance, Hsieh and Urquiola (2006) report that in Chile competition is not associated with higher public school performance because, according to the authors, private voucher schools respond to the competitive pressures created by the voucher system not by raising productivity, but rather by choosing better students. Conversely, Gallego (2005) finds that greater competition is associated with higher student achievement. In addition, most studies comparing standardized test scores between private voucher and public schools show a private school advantage, although the differences are usually small (Anand et al., 2009; Sapelli and Vial, 2002; McEwan, 2001).

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the private voucher sector does not necessarily imply that all private school types are segregated by social class. Most of the existing literature in Chile treats private schools as an aggregate category where all schools are considered to be largely identical, when in fact there is variation across private voucher school sectors. For instance, while many non-profit voucher schools (religious and secular), which enroll 16 percent of students, declare in their missions that they target low-income and minority students, for-profit schools, which represent one-third of enrollments, are profit oriented and more likely to target more advantaged and less expensive to educate students.

In addition, when considering how much educational vouchers and private schools contribute to the segregation of minorities and children from low-income families, there are two components to the question. One is between sector segregation – the proportion of disadvantaged students enrolled in a sector – and the other is between school segregation within a sector – the isolation of disadvantaged students within a sector. For example, although private schools enroll fewer disadvantaged students than public schools in Chile, they may be evenly distributed across all private schools, while poor and minority students may be isolated from their more advantaged counterparts in public schools. Therefore, while the between-sector segregation may be more pronounced in private schools, the between-school segregation may be greater in public schools.

While in the past, Chilean researchers and policymakers have focused on the extent of stratification between public and private schools (Hsieh and Urquiola, 2006; OECD, 2004), segregation within a sector may have a more adverse effect on educational outcomes than between-sector stratification. For example, researchers have found that high levels of between school segregation within a sector can have a negative effect on the learning outcomes of disadvantaged students that attend the most segregated schools (Zimmer and Toma, 2000; Hoxby, 2002; Kahlenberg, 2001; Hanushek et al., 2009).² School segregation is also related to other outcomes. For instance, Mayer (1991) finds that attendance at a segregated high school increases teen pregnancy and the probability of dropping out (also see, Guryan, 2004). Research evidence also suggests that school segregation within a sector can have a negative impact on non-cognitive outcomes – attitudes and behaviors – that are important to employers (Crain and Strauss, 1985; Rothstein, 2004; Wells et al., 2008). Scholars also find that students who experience the most racial and ethnic segregation in classroom settings score the lowest on intellectual engagement, motivation, and citizenship indices (Levinson and Levinson, 2003).³ School socio-economic segregation within a sector can also have a negative effect on the educational environment at a school. For example, Hanushek et al. (2004) find that high school poverty rates contribute to teacher turnover and to a school's ability to attract experienced teachers. Thus, when considering how much private school vouchers contribute to segregation, researchers and policymakers should consider both segregation between sectors and within a sector.

Using a highly detailed panel dataset constructed from the administrative records of the Ministry of Education, I examine public and private voucher school socio-economic and indigenous segregation. I find that public schools are more likely to serve disadvantaged student populations than private voucher schools. I also find that disadvantaged students are more segregated in the

private voucher sector than in public sector. These patterns are constant over time. While between and within sector socio-economic and indigenous segregation levels vary across private school types, the differences are not always consistent with theory. The data also suggest that school tuition is much more important than other factors in explaining socioeconomic segregation levels between private voucher school sectors.⁴ I also find preliminary suggestive evidence that an adjusted voucher policy could mitigate the stratifying effects of educational vouchers.

The remainder of the paper is organized as follows. The next section briefly reviews the literature on school choice and stratification. The second section presents some background on Chile's voucher program and describes the school types that will serve as the analytical categories. Section three examines between-sector segregation across public and private voucher schools in Chile. The fourth section analyzes between-school socio-economic and indigenous segregation within public and private voucher school sectors. Section five explores the potential impact of policies (shared financing and an adjusted voucher) on socioeconomic and indigenous segregation. The final section concludes and discusses policy implications.

2. The literature: school choice and stratification

Discussions of the effects of school choice and vouchers have often raised questions about social equity. Viteritti (2003) maintains that the significant question in school choice is not an empirical one, rather it is a normative question: whether or not to extend school choice to a population that does not enjoy its advantages. He argues that an educational system that provides choice to some and not others is inherently unfair, especially when the opportunity is determined by income. Since the option of school choice through residential mobility or through enrollment in private schools has long been available to more wealthy families, Viteritti (2003) and other voucher proponents maintain that expanding the right of disadvantaged parents to leave their low performing neighborhood schools for higher performing ones may improve social equity as parental income becomes less important in determining who attends higher quality private schools (Neal, 2002; Sugarman, 1999). Voucher advocates have also argued that given the high levels of residential segregation in most metropolitan areas, choice can reduce school segregation by providing disadvantaged parents access to more integrated schools (Moe, 2001).

As the push for school choice has intensified in many countries (Plank and Sykes, 2003), a series of critiques have identified the costs of educational vouchers. Perhaps the most consistent criticism is that choice increases the risk of school segregation. Voucher critics have frequently argued that disadvantaged parents do not have the ability to make good choices. Ascher et al. (1996) argues that parents are not “natural consumers of education” and that few parents are able to acquire the information necessary to make informed educational choices. Similarly, Henig (1994) maintains that low-income and minority parents will base their decisions on non-academic aspects of education, including the availability of sports and other extracurricular activities, and the proximity of schools to home or work. Critics are concerned that if minority and low-income families choose schools based on non-educational dimensions and white and wealthy families focus their choices on academic quality, then schools will become more segregated, as more advantaged families choose higher quality schools, leaving disadvantaged children in segregated, low-quality schools.

Other skeptics are concerned that parental choice may increase diversity among academic programs across schools in response to different tastes in educational philosophies, and this may increase

² There is also some evidence that the proportion of low achieving peers has a moderately negative effect on the performance of regular students (Imberman et al., 2009; Lavy et al., 2008). Some researchers also find that the effects are less pronounced once differences in the prior achievement of students are accounted for (Hoxby and Weingarth, 2005).

³ See Massey and Denton (1993), Fischer and Massey (2006), and Camille et al. (2004) on the negative effects of residential segregation on academic and other outcomes.

⁴ Public schools are open to all students and cannot charge tuition.

segregation. For instance, working class families may choose schools that emphasize traditional childrearing values and upper-middle class families may choose schools that emphasize abstract thinking and the development of problem solving skills (Lareau, 2003). In the Netherlands, for instance, researchers have found that private subsidized schools that adopt a specific teaching method (e.g. Montessori) are more likely to serve white middle class families, while Islamic and Hindu subsidized schools are more likely to serve low-income minority students (Karsten et al., 2006). Researchers in the United States report that charter schools⁵ that emphasize an Afro centrist curriculum are more likely to attract African American students than whites and charter schools that emphasize a bilingual curriculum are more likely to draw Hispanic students (Fuller, 2000).

Others have suggested that parents will select schools based on the race or class composition of their student bodies and not on their academic programs (Jellison, 2002; Saporito and Lareau, 1999; Wells, 1996). Most studies in the United States find that parents prefer and choose schools with children of the same race or socioeconomic status (Frankenberg et al., 2010; Saporito, 2003; Buckley and Schneider, 2002; Weiher and Tedin, 2002). However, research on the voucher programs in Washington, DC (Greene and Winters, 2006) and Cleveland (Forster, 2006), and suggests that private voucher schools are less segregated than traditional public schools. Though more recent evidence shows that the Milwaukee voucher program is neutral in its effect on racial segregation (Greene et al., 2010). Research in Germany shows that Turkish children are more likely than German children to attend a school with a larger proportion of foreign students (Kristen, 2008). Researchers have also found evidence of 'white flight' and 'Latino flight' from public schools into private schools when public schools have large concentrations of blacks or minorities (Farlie, 2002; Clotfelter, 1976, 2001) and 'native flight' from public schools in response to inflows of immigrants (Betts and Farlie, 2003). Recent evidence on the Netherlands' national voucher program shows that ethnic and socioeconomic school segregation levels are high both absolutely and relative to those in cities in the United States (Ladd et al., 2009).

Most empirical research on Chile's universal voucher program also confirm school choice skeptics' concerns. For instance, a recent study in Chile finds that parents are more likely to choose schools based on the class composition of the student body than on objective academic performance (Elacqua et al., 2006). Researchers have also found that Chile's unrestricted flat per-pupil voucher has led to increased stratification across schools. For example, Hsieh and Urquiola (2006) find that the main effect of unfettered school choice in Chile was the flight of middle class students from public schools. In municipalities where the private voucher school sector grew, the authors show that there was a greater decline in the socioeconomic status of public school students relative to the municipality average.

While voucher critics may take this as evidence that unregulated voucher programs will lead to greater socio-economic segregation in the private sector, it may be unwise to generalize the results to all private schools. Much of the existing empirical literature in Chile treats private schools as an aggregate category and few if any studies have examined whether socio-economic segregation levels differ across for-profit and non-profit (religious and secular) voucher schools.⁶

There is a vigorous scholarly debate on the comparative behavior of for-profit and non-profit schools. Choice skeptics have argued that for-profit schools will try to save costs by recruiting and retaining students who are least demanding in terms of resources (Levin, 2002). Some scholars have also argued that non-profits are better positioned to provide high quality education to disadvantaged students than for-profit schools because they may rely on donations of money or volunteer time to finance the higher costs involved in educating poor and minority students, while for-profits must survive in the education marketplace (Rose-Ackerman, 1996). Some have questioned whether or not there are likely to be systematic differences in the objectives of for-profit and non-profit schools. For example, non-profits may be highly selective and, conversely, for-profit schools may have a deep commitment to educating disadvantaged students.⁷

Empirical research has identified significant differences in for-profit and non-profit behaviors in many mixed industries, including hospitals, nursing homes, prisons, day care centers, and various social services. For instance, Lukesetich et al. (2000) finds that non-profit nursing homes spend more per-patient on nursing care and less on administrative expenses than for-profit nursing homes. In studies of hospitals, Schlesinger et al. (1987) finds that non-profit hospitals provide more uncompensated care than for-profits. The empirical studies of day care centers also show that non-profits provide higher quality services for lower fees (Morris and Helburn, 2000). Hart et al. (1997) finds that non-profit prisons hire higher quality prison guards than for-profits. Some researchers have made finer distinctions between non-profit charter schools. For instance, recent research distinguishes between "market-oriented" and "mission-oriented" non-profit charter schools (Brown et al., 2005). Although they find differences in the levels of racial and socio-economic segregation across market-oriented schools and mission-oriented schools, the theoretical typology they use for distinguishing between non-profit charter schools limits their ability to disentangle the inter-institutional differences driven by incentives and legal constraints placed on non-profit and for-profit schools.⁸

The evidence on this point is limited because most schooling systems do not provide public funding to for-profit schools. In most countries schools continue to be funded and operated primarily by the government (OECD, 2009) and non-profit status is usually required for private schools (James, 1993). Researchers can gain insight into this debate through the examination of school segregation patterns across socioeconomic and minority groups in Chile's mixed educational market in which public, private non-profit (religious and secular), and for-profit voucher schools compete for enrollments.

The issue of for-profit schooling was one of the main factors that sparked one of the largest protests in Chilean history. In July of 2006, more than 600,000 students walked out of class and occupied hundreds of schools across the country, demanding that the government cease funding for-profit schools (El Mercurio, 2006). Students and others opposed to funding for-profit schools argued that for-profit providers could not be trusted to place the interests of disadvantaged children ahead of profitability (OEI, 2007). Critics countered that for-profit schools expand the diversity of private schooling options for many low-income students in Chile (Libertad y Desarrollo, 2007). Neither of these arguments, however, is based on any data in Chile on student

⁵ Charter schools are independent schools designed and operated by teachers, parents, community leaders, educational entrepreneurs, and others. They are sponsored by local or state educational organizations who monitor their quality and effectiveness but allow them to operate outside of the traditional public school system. See http://www.nps.k12.va.us/NCLB/NCLB_glossary.htm.

⁶ See McEwan (2001) for an analysis of student achievement across Catholic and public schools in Chile.

⁷ Weisbrod (1998) finds that mission driven organizations often consider the constraints placed on non-profits too restrictive, and profit maximizing firms often find it more advantageous to choose non-profit status due to tax exemptions.

⁸ Institutional features include distributional constraints, tax exemptions, and private donation laws (Weisbrod, 1998).

socio-economic segregation across the different types of private schools (for-profit or non-profit).

One goal of this paper is to inform the scholarly and education policy debate in Chile by examining socio-economic patterns of school attendance across for-profit, non-profit, and public schools.

3. Background on Chile

During the 1980s, the military government enacted a sweeping education reform program (1973–1990). First, the Ministry of Education decentralized education service delivery to regional and provincial offices and the administration of public schools to municipal governments, whose maximum authority is an elected mayor. Second, the government altered the financing scheme of public and most private schools. Municipalities and private school owners that did not charge tuition started to receive a flat per-pupil voucher.⁹ As a result, enrollment gains or losses began to have an impact on their budgets. Fee-charging private schools continued to operate without public funding. The essential features of this system have remained in place for almost three decades (OECD, 2004; Cox, 2003).

The provision of education in Chile has become increasingly privatized since the voucher reforms were instituted. In 1981, 15 percent of Chilean K-12 students attended private schools that received some public subsidy, and another 7 percent attended more elite, unsubsidized private schools. Between 1981 and 1986 more than 1000 new schools entered the education market and the private voucher enrollment rate increased from 15 percent to 25 percent. By 1990, over 31 percent of students attended private voucher schools. Between 1990 and 2006 the enrollment rate in private voucher schools increased from 31 percent to 43 percent of total enrollments. Most of these gains were at the expense of public school enrollments. Adding in the 6 percent of students in elite private non-voucher schools, leaves a slight majority of Chilean students in public schools.

Most researchers generally use a single category to describe private voucher schools in Chile. However, there is a great deal of variability in the private voucher sector. Prior to the voucher reforms in 1981, most subsidized private schools were non-profit (Aedo, 2000). When private subsidized schools began to receive the same per-pupil payment as the public schools, a number of mostly for-profit voucher schools entered the market. Tables 1 and 2 show how primary school students are distributed across school types. Public schools account for the majority (59 percent) of schools and less than half of enrollments (49 percent). For-profit voucher schools account for 27 percent of schools and 29 percent of enrollments. For-profit schools are especially small in size and scale (only 20 percent belong to a network), suggesting that when these groups set out to establish a school, they are probably looking to create a school that only provides services to children in the community.¹⁰ Non-profit voucher schools account for about 9 percent of schools and 15 percent of enrollments. These schools, which are often subsidized by the Church (72 percent are Catholic) or local businesses, often have access to donated facilities and staff willing to work for below-market salaries, and thus are able to provide a range of services to disadvantaged students whose costs exceed the voucher and tuition payments.

⁹ Chile's voucher formula includes adjustments for rural schools and high schools, but until very recently did not take into account student socioeconomic characteristics or the existence of a high concentration of low-income students in public and private voucher schools.

¹⁰ Recent survey evidence suggests that most for-profit voucher school owners are former public school teachers (Corvalan et al., 2008).

4. Between sector stratification

To examine enrollment practices across public and private voucher schools, I use a highly detailed original panel dataset I have constructed from the administrative records of the Ministry of Education of Chile. I will use two measures to compare the level of stratification between sectors. The first measure is the vulnerability index (IVE). IVE is constructed by the National Scholarship and School Aid Board (JUNAEB). JUNAEB estimates IVE on the basis of the results of a parent survey conducted by schools. These surveys provide information about the student's socioeconomic background. The IVE has a minimum value of 0, which represents 0 percent of children at social risk, and goes up to a maximum value of 100, indicating the most socially disadvantaged schools (100 percent social risk). I acknowledge that using this indicator as a proxy for low socioeconomic status is not perfect.¹¹ I chose this index for three reasons. First, the Ministry of Education uses IVE to determine eligibility for school breakfast and lunch programs. Second, qualifications for the program are dependent on the family demonstrating low levels of income and education and other social indicators (e.g. health and hygiene). Third, JUNAEB has published comparable IVE data since 2000.¹²

The second measure I use to examine the level of indigenous segregation between and within public and private school sectors is the percentage of indigenous students enrolled across sectors. I used parent questionnaires collected by Chile's national standardized test (*Sistema de Medicion de la Calidad de la Educacion-SIMCE*) to estimate the indigenous student population in schools. The survey question asks parents if the child's mother identifies herself as indigenous.¹³ The proportion of indigenous students is similar to the overall population (Mideplan, 2007).

Tables 3 and 4 present socio-economic and indigenous patterns of school attendance across school types. The data confirm prior research suggesting that, in the aggregate, private voucher schools serve a lower percentage of disadvantaged (low socioeconomic status and indigenous) students than public schools. However, the results also indicate that this aggregate analysis masks some differences within the private voucher sector. For instance, Catholic schools serve a smaller percentage of disadvantaged students than for-profit and Protestant schools. Table 3 also indicates that indigenous students make up 8.1 percent of for-profit enrollments and only 6.4 percent of non-profit enrollments. Indigenous student enrollment rates in Protestant schools (9.5 percent in 2006) are similar to those of public schools.

5. Between school stratification

Although the previous results indicate that a higher percentage of disadvantaged students (low socioeconomic status and indigenous) are enrolled in public schools as well as some differences across private voucher school sub-sectors, they do not indicate how these students are distributed within each sector. The public sector in Chile could, for example, be highly segregated with the poor isolated from their middle and high-income counterparts.

¹¹ Croxford (2000) and Valenzuela et al. (2008) discuss some of the drawbacks of this measure. In a previous iteration of this research, I included an analysis of mothers' level of education (elementary school and less than high school) as a proxy for social disadvantage. Here I only report IVE, but my results are substantively similar when I replace IVE with mother's level of education. These results are available upon request.

¹² JUNAEB has published IVE scores since 1996. However, due to changes in the methodology used to construct this measure, IVE school level data are only comparable between 2000 and 2006 (Marshall and Correa, 2005).

¹³ Specifically the question asks whether the student's mother belongs to one of the following nine indigenous groups: Aymara, Rapa-Nui, Quechua, Mapuche, Atacamenno, Coya, Kawaskar, Yagan, and Diaguita.

Table 1

Distribution of primary schools across school types, 1999–2006.

	1999	2000	2001	2002	2003	2004	2005	2006
Public	64.5%	63.8%	63.1%	62.4%	61.3%	60.8%	60.3%	59.3%
For-profit voucher	19.7%	20.4%	21.4%	22.2%	23.6%	24.6%	26.2%	27.0%
Independent	16.3%	16.7%	17.4%	18.1%	19.1%	19.9%	21.4%	22.4%
Network	3.4%	3.7%	4.0%	4.2%	4.5%	4.7%	4.7%	4.6%
Non-profit voucher	8.0%	8.1%	8.1%	8.4%	8.5%	8.4%	8.3%	8.7%
Catholic	4.7%	4.7%	4.8%	4.9%	5.0%	4.9%	4.9%	5.1%
Protestant	0.9%	0.9%	0.9%	1.0%	0.9%	0.9%	0.9%	1.0%
Secular	2.4%	2.4%	2.4%	2.5%	2.6%	2.6%	2.5%	2.6%
Private non-voucher	7.9%	7.7%	7.5%	7.1%	6.6%	6.2%	5.2%	5.1%
Total	100%	100%	100%	100%	100%	100%	100%	100%
Number of schools	8,723	8,768	8,834	8,871	8,953	8,958	8,956	8,899

Table 2

Distribution of primary school students across school types, 1999–2006.

	1999	2000	2001	2002	2003	2004	2005	2006
Public	57.1%	56.4%	55.7%	54.5%	53.4%	52.3%	51.3%	49.7%
For-profit voucher	21.3%	22.0%	22.8%	23.8%	25.0%	26.2%	28.0%	29.1%
Independent	16.0%	16.2%	16.7%	17.3%	18.4%	19.4%	21.2%	22.4%
Network	5.3%	5.9%	6.1%	6.4%	6.6%	6.9%	6.8%	6.7%
Non-profit voucher	13.6%	13.7%	13.7%	14.0%	14.3%	14.3%	14.4%	14.8%
Catholic	10.4%	10.4%	10.4%	10.7%	10.8%	10.9%	11.0%	11.3%
Protestant	1.3%	1.3%	1.3%	1.3%	1.4%	1.4%	1.4%	1.4%
Secular	2.0%	2.0%	1.9%	2.0%	2.1%	2.0%	2.1%	2.1%
Private non-voucher	7.9%	7.8%	7.8%	7.7%	7.4%	7.2%	6.3%	6.3%
Total	100%	100%	100%	100%	100%	100%	100%	100%
Number of students	2,322,289	2,362,356	2,366,159	2,349,165	2,326,991	2,279,040	2,234,323	2,189,561

Similarly, indigenous–non-indigenous segregation may be greater in the public sector than in the private voucher sector.

Researchers in the United States have examined between school racial and ethnic segregation within public and private school sectors. For instance, [Coleman et al. \(1982\)](#) find that although private schools in the United States tend to enroll fewer black students than public schools, black–white segregation is lower among private schools than among public schools. [Clotfelter \(2004\)](#) reports similar findings. [Bryk et al. \(1993\)](#) find that Catholic schools are less segregated by race and socioeconomic status than public schools and secular private schools. More recently, [Betts and Farlie \(2001\)](#) find that Hispanic–white segregation is greater in the public sector than in the private sector.

There is an extensive debate in the social science literature on the most appropriate way to measure school segregation ([Taeuber and James, 1982](#); [Massey and Denton, 1993](#)). The vast majority of applied work uses measures of evenness and exposure ([Echenique and Fryer, 2007](#)). Evenness refers to the differential distribution of students across schools. Exposure measures the amount of contact between different groups of students (e.g. by race, social class) across schools.¹⁴ To examine how disadvantaged students are distributed among public and private schools in Chile, I use the dissimilarity index, a measure of evenness. The dissimilarity index is designed to approximate the extent to which disadvantaged students are unevenly distributed across schools. It can also be interpreted as the percentage of students that would have to change schools in order to produce a completely integrated distribution of students.¹⁵

The dissimilarity index has at least one undesirable feature. It does not conform to the principle of transfers. That is, school segregation is not reduced if disadvantaged students move from schools of higher to lower proportions of disadvantaged students or if more advantaged students move in the opposite direction. The dissimilarity index is only unambiguously reduced if both schools are moved toward the zero segregation condition. Other commonly used measures of school segregation, such as measures student “isolation”, possess this property. However, the dissimilarity index has the appealing and unique feature among segregation measures – which is important for the present analysis – that it is invariant to the size of the disadvantaged student group. This allows researchers to compare levels of school segregation across countries, cities, and years.¹⁶

I calculate dissimilarity indices for public schools and private voucher schools, for-profit and non-profit voucher schools, and for different sub-sectors of for-profit and non-profit voucher schools.¹⁷ By this measure of segregation, [Tables 5 and 6](#) indicate that among all sectors, public schools have the most diverse enrollments and for-profit independent schools have the highest levels of socio-economic and indigenous segregation. I also find that socioeconomic segregation is lower among Protestant and Catholic schools than among for-profit and secular schools, but indigenous–non-indigenous segregation is greater in religious schools than in for-profit schools that belong to a network. [Table 5](#) also indicates that socioeconomic segregation declined slightly

¹⁴ [Massey and Denton \(1988\)](#) consider other measures of segregation that are less common in education policy research. For example, they describe concentration indices, which measure the relative amount of physical space occupied by disadvantaged groups. They also consider centralization indices which measure the extent to which a group is located near the center of a city.

¹⁵ The findings I report across subsectors of public and private voucher schools are robust to the use of different segregation indices of evenness (segregation index) and exposure (isolation index). These results are available upon request.

¹⁶ See [Frankel and Volij \(2009\)](#), [Echenique and Fryer \(2007\)](#), and [Massey and Denton \(1993\)](#) for a technical discussion of the dissimilarity index and other approaches to measuring segregation.

¹⁷ I have excluded rural students from this analysis because most students who attend these small schools, most of which are public, are disadvantaged (and thus highly segregated) and their parents have few school choice opportunities. The results, nonetheless, are very similar when I include rural schools in the analysis. These results are available upon request. I have also excluded private non-voucher schools from this analysis because they enroll students mainly from high income families (see [Tables 3 and 4](#)).

Table 3

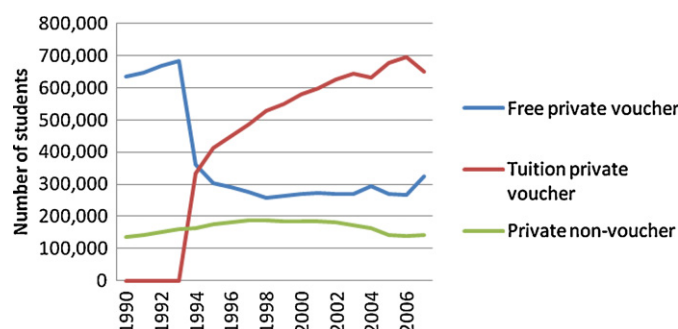
Vulnerability index across public, for-profit, and non-profit voucher schools (2000–2006).

	2000	2001	2002	2003	2004	2005	2006
Public	39.1%	41.8%	39.8%	39.9%	39.7%	39.3%	39.6%
Private voucher	20.2%	21.6%	20.6%	21.4%	20.6%	19.7%	20.5%
For-profit voucher	20.9%	22.9%	21.9%	22.4%	21.8%	21.4%	21.5%
Independent	21.9%	24.1%	22.9%	22.3%	21.6%	21.4%	21.3%
Network	18.5%	20.3%	19.5%	22.7%	22.3%	21.5%	22.1%
Non-profit voucher	18.9%	19.3%	18.3%	19.6%	18.8%	18.4%	19.2%
Catholic	18.3%	18.6%	17.6%	18.9%	17.9%	17.6%	18.5%
Protestant	20.8%	20.9%	20.5%	21.7%	21.8%	21.2%	22.8%
Secular	22.3%	23.2%	21.8%	24.0%	23.1%	22.2%	21.0%
Private non-voucher	0.3%	0.7%	0.7%	0.9%	1.0%	0.0%	0.0%

Table 4

Indigenous students across public, for-profit, and non-profit voucher schools (2000–2006).

	1999	2006
Public	8.1%	9.6%
Private voucher	6.0%	7.4%
For-profit voucher	6.4%	8.1%
Independent	6.7%	8.2%
Network	5.7%	7.7%
Non-profit voucher	5.2%	6.4%
Catholic	5.0%	6.1%
Protestant	7.3%	9.5%
Secular	4.5%	5.2%
Private non-voucher	0.7%	0.6%

**Fig. 1.** Number of public and private voucher free and tuition, and non-voucher students, 1990–2006.**Table 5**

High risk (vulnerable) and low risk student dissimilarity index values for public, for-profit and non-profit voucher urban schools, 2000–2006.

	2000	2001	2002	2003	2004	2005	2006
Public and private voucher	0.35	0.38	0.36	0.35	0.36	0.37	0.35
Public	0.25	0.28	0.26	0.23	0.24	0.25	0.23
Private voucher	0.43	0.45	0.42	0.43	0.44	0.45	0.42
For-profit	0.45	0.47	0.44	0.46	0.45	0.46	0.44
Independent	0.45	0.47	0.45	0.47	0.46	0.48	0.46
Network	0.46	0.47	0.43	0.42	0.40	0.43	0.40
Non-profit	0.39	0.41	0.38	0.39	0.40	0.39	0.37
Catholic	0.39	0.40	0.37	0.38	0.40	0.39	0.36
Protestant	0.39	0.41	0.37	0.33	0.37	0.35	0.33
Secular	0.44	0.47	0.47	0.45	0.41	0.41	0.42

over time in all school sectors, with the exception of for-profit independent schools. Conversely, indigenous student segregation levels reported in Table 6 have increased within the public, for-profit independent, Protestant, and secular non-profit school sectors and remained stable between 1999 and 2006 within the Catholic and for-profit network sectors.

6. Public policies and stratification

The preceding sections have focused on segregation between and within the public and private school sectors. In this section, I turn my attention to the impact of two different policies – both designed to change the rules under which the Chilean voucher system operates – on socio-economic and indigenous segregation. First, I analyze the effect of the change in the school financing mechanism that was instituted in 1994. Under this rule, private voucher schools can charge monthly tuition fees in addition to the voucher. Second, I analyze the effect of the targeted adjusted voucher program (*Ley SEP*) on school socio-economic and indigenous segregation. This law provides a larger per-student

voucher for schools serving students with low socioeconomic status.

6.1. Shared financing

In 1994, the Ministry instituted a shared financing scheme that allowed all private voucher schools—both elementary and second-

Table 6

Non-indigenous and indigenous student dissimilarity index values for public and for-profit and non-profit voucher urban schools, 1999 and 2006.

	1999	2006
Public and private voucher	0.40	0.42
Public	0.36	0.39
Private voucher	0.43	0.44
For-profit	0.42	0.43
Independent	0.44	0.45
Network	0.38	0.38
Non-profit	0.44	0.44
Catholic	0.44	0.44
Protestant	0.38	0.42
Secular	0.36	0.42

Table 7

Vulnerability index across public and private free and tuition voucher schools, 2000–2006.

	2000	2001	2002	2003	2004	2005	2006
Public	39.1%	41.8%	39.8%	39.9%	39.7%	39.3%	39.6%
Private voucher free	35.4%	38.5%	37.3%	38.6%	35.4%	36.4%	37.0%
For-profit free	39.3%	43.2%	42.3%	43.6%	41.3%	42.8%	42.4%
Non-profit free	30.5%	32.5%	30.8%	32.4%	31.5%	30.5%	30.6%
Private voucher tuition	13.1%	13.9%	13.4%	14.2%	13.7%	13.0%	14.2%
For-profit tuition	13.8%	15.1%	14.4%	15.0%	14.6%	14.3%	14.9%
Non-profit tuition	11.8%	11.5%	11.4%	12.5%	11.5%	11.8%	13.2%
Private non-voucher	0.3%	0.7%	0.7%	0.9%	1.0%	0.0%	0.0%

Table 8

Indigenous students across public and private free and tuition voucher schools, 1999 and 2006.

	1999	2006
Public	8.1%	9.6%
Private voucher free	11%	15.3%
For-profit free	12%	18.5%
Non-profit free	8.9%	11.5%
Private voucher tuition	3.6%	4.4%
For-profit tuition	4.0%	4.8%
Non-profit tuition	3.0%	3.8%
Private non-voucher	0.7%	0.6%

Table 9

High risk (vulnerable) and low risk student dissimilarity index values For public and private free and tuition voucher urban schools, 2000–2006.

	2000	2001	2002	2003	2004	2005	2006
Public	0.25	0.28	0.26	0.23	0.24	0.25	0.23
Private voucher free	0.35	0.38	0.34	0.31	0.33	0.32	0.31
For-profit free	0.39	0.41	0.35	0.32	0.32	0.32	0.32
Non-profit free	0.29	0.33	0.29	0.27	0.27	0.26	0.26
Private voucher tuition	0.42	0.43	0.40	0.44	0.44	0.46	0.42
For-profit tuition	0.43	0.44	0.42	0.46	0.45	0.46	0.43
Non-profit tuition	0.39	0.39	0.37	0.39	0.41	0.41	0.36

Table 10

Non-indigenous and indigenous student dissimilarity index values for public and private free and tuition voucher urban schools, 1999 and 2006.

	1999	2006
Public	0.36	0.40
Private voucher free	0.44	0.44
For-profit free	0.44	0.44
Non-profit free	0.42	0.43
Private voucher tuition	0.38	0.40
For-profit tuition	0.37	0.40
Non-profit tuition	0.39	0.39

Table 11

High risk (priority) and low risk student dissimilarity index values For public and private voucher urban schools, 2007 and 2008.

	2007	2008
Public and private voucher	0.35	0.34
Public	0.26	0.25
Private voucher	0.36	0.34
For-profit	0.34	0.33
Independent	0.36	0.34
Network	0.30	0.29
Non-profit	0.35	0.34
Catholic	0.34	0.32
Protestant	0.34	0.31
Secular	0.42	0.38

Table 12

Non-indigenous and indigenous student dissimilarity index values for public and private voucher urban schools, 2007 and 2008.

	2007	2008
Public and private voucher	0.43	0.42
Public	0.41	0.40
Private voucher	0.44	0.43
For-profit	0.42	0.43
Independent	0.43	0.45
Network	0.37	0.37
Non-profit	0.46	0.43
Catholic	0.45	0.42
Protestant	0.43	0.39
Secular	0.40	0.40

ary and public secondary schools to charge limited tuition (Montt et al., 2006). The “shared financing” law in Chile allows private voucher schools and public high schools to charge monthly fees, in addition to the per-pupil subsidy, that can be up to 1.6 times the basic voucher payment.¹⁸ In Chile, 50 percent of private voucher schools charge some tuition. Fig. 1 illustrates that private voucher schools that charge tuition have far outpaced growth rates of free private voucher schools since 1994.

Tables 7 and 8 highlight the stark differences in disadvantaged (low socioeconomic status and indigenous) student enrollment patterns between free and tuition charging private voucher schools. There is also significant variation across free private voucher school sub-sectors. For instance, free for-profit schools serve a larger proportion of disadvantaged students than public schools. Conversely, free non-profit schools serve a smaller percentage of students with low socioeconomic status than free for-profit and public schools. However, they enroll a higher proportion of indigenous students than public schools. The differences between for-profit and non-profit voucher schools that charge tuition are less significant.

The dissimilarity index reported in Table 9 demonstrates that students of low socioeconomic status are significantly more segregated in private voucher schools that charge tuition than in free private voucher schools. This is suggestive evidence¹⁹ that while public and free for-profit private voucher schools are non-selective, tuition-charging private voucher schools select student populations based on ability to pay. However, the findings reported in Table 10 also suggest that school fees may not be as important in explaining indigenous student segregation patterns. For instance, according to Table 10, indigenous students are less segregated in for-profit and non-profit schools that charge tuition than indigenous students enrolled in free private voucher schools.

¹⁸ Discounts to vouchers are applied progressively. If monthly tuition is less than half the level of the School Voucher Unit (USE), no discount is applied. Tuition fees between one half and one USE incur a 10 percent deduction. Fees between one and two USE incur a 20 percent deduction. Fee charging schools must also devote up to 10 percent of their additional income to finance scholarships.

¹⁹ The results of this analysis do not ascribe a causal relationship, which would require school-level socioeconomic data prior to the implementation of the shared financing policy in Chile. These data are currently not available.

6.2. Adjusted voucher

In 2008, the Chilean legislature enacted the adjusted voucher law (*Ley de Subvención Preferencial or SEP*). The SEP law recognizes that it is more costly to educate students with low socioeconomic status by introducing an extra per-pupil subsidy (50 percent over the base voucher) for students classified as priority in the Ministry of Education's socioeconomic status classification system and for schools with a high concentration of priority students.²⁰ The additional per-student voucher is tied to an increased role of the Ministry of Education in monitoring and classifying schools based on student performance and holding them accountable for their outcomes (Elacqua et al., 2009). The SEP law also forbids participating schools from using parental interviews and admissions tests to select and expel students. In addition, participating schools cannot charge tuition to priority students.

Tables 11 and 12 show changes in student socio-economic segregation levels of students classified as priority and indigenous segregation levels across all subsidized schools (public and private voucher) before and after the SEP law was instituted in 2008. The results indicate that, between 2007 and 2008, socioeconomic segregation declined slightly across all school sectors and indigenous segregation fell across public and non-profit school sectors. This is not to downplay the segregation that still persists at fairly high levels in most sectors. But this preliminary evidence suggests that the adjusted voucher program may provide schools with incentives to enroll more disadvantaged student populations and reduce socio-economic and indigenous segregation. More research needs to be conducted as more data becomes available to determine whether or not these trends will hold over time.

7. Conclusion and policy implications

Consistent with previous research, I find that public schools are more likely to serve disadvantaged (low socioeconomic status and indigenous) student populations than private voucher schools. I also find that disadvantaged students are less segregated in the public sector than in the private voucher sector. These results are not surprising given that public schools are mandated by law to accept all students who apply, regardless of ability to pay, while private schools are permitted to use parental interviews to select and expel students as they see fit. I also find evidence that these aggregate patterns may be masking some differences across private voucher school sectors. For-profit schools, surprisingly, are more likely to serve disadvantaged students than non-profit voucher schools. However, disadvantaged students in for-profit schools are more isolated from their more advantaged peers than in non-profit and public schools. This suggests that this sub-sector of schools is finding market niches in both low and middle-income communities.

Perhaps the most counter-intuitive finding presented in this the paper is that, despite having a mission to serve the needy, Catholic voucher schools enroll, on average, fewer disadvantaged students (low socioeconomic status and indigenous) than public and other private voucher school types. This pattern is likely due to several factors. First, Catholic schools may have been forced to adapt their personnel practices to meet the needs of the current Chilean workforce. For instance, in the past, Catholic schools were able to rely on cheap labor costs. However, according to survey evidence in Chile, the number of women and men currently going into religious life has dwindled (Conferencia Episcopal de Chile, 2002). At

present, just 4 percent of the professional staff at Catholic schools is religious and clergy. Lay teachers who replaced the religious teachers may be demanding higher wages. As labor costs rise, Catholic schools in Chile may be serving fewer disadvantaged students to account for higher costs.

A second reason why Catholic schools may be focusing on a more elite clientele may be related to what parents value when they choose a school. For instance, recent evidence on the Chilean voucher system suggests that parents care about student demographics when choosing a school (Elacqua et al., 2006). In Chile, private voucher schools are given latitude to levy fees and select students from more high status families and refuse applicants from disadvantaged families. Thus, if parents care about demographics, schools will be less inclined to enroll disadvantaged students.

Results from this study have implications for future research and policy. The findings reported here demonstrate that, when considering how much private school vouchers contribute to segregation, it is important to examine both between-sector segregation and between-school segregation within a sector. The findings also suggest that education research should not treat private schools as an aggregate category where all schools are considered to be largely identical.

From a policy perspective, the results of this study suggest that private schools respond differently to vouchers than public schools. It also suggests that policies that provide schools with incentives to charge tuition, even if it is only limited, can have a negative impact on school segregation.²¹ However, the preliminary findings also suggest that an adjusted voucher that provides an extra per-pupil subsidy for disadvantaged students can mitigate some of the stratifying effects of unfettered voucher programs.

Overall, segregation remains moderately high in Chile,²² but there has been a slight decline of socioeconomic school segregation over the last decade. However, more research needs to be conducted to determine the factors that influence or reduce student segregation. For instance, how important is the voucher design? Most studies suggest that a flat per-pupil voucher leads to greater segregation across schools (Epple and Romano, 1998). Preliminary evidence presented in this paper suggests that the socioeconomic segregation decline between 2007 and 2008 may have been linked to the adjusted voucher instituted in 2008. If the voucher was differentiated by student ethnicity, would schools be willing to enroll more indigenous students? If the adjusted voucher was greater than 50 percent, would it create larger markets for private voucher schools in low-income communities? How does private school screening affect stratification? The ability of private schools in Chile to select students undoubtedly leads to greater student segregation. Sorting is also likely induced by the demand side factor of self-selection, particularly in tuition charging schools.

Disentangling the effects of school location decisions and demographic changes on school segregation also remain an

²¹ There is some anecdotal evidence that the shared financing policy may have helped foster school integration in elite private non-voucher schools. Between 2002 and 2006, 30 percent of elite non-voucher schools closed down. Many of these schools re-opened as tuition charging private voucher schools and began to enroll students from diverse social backgrounds (Hales, 2009).

²² International comparisons suggest that school segregation in Chile is not higher, and is in some cases lower, when compared to school segregation – using the index of dissimilarity – in other developing and industrialized countries. For instance, Chilean public school students appear significantly less segregated than British students when the IVE or free school meal eligibility is used as a proxy for social disadvantage (Allen and Vignoles, 2006). And segregation of indigenous students in public and private voucher schools in Chile is distinctly lower than indigenous students in Bolivian schools (McEwan, 2004), black students of African or Caribbean heritage and Indian and Pakistani students in England's schools (Burgess and Wilson, 2004), and African-American students in schools in the United States (Taeuber and James, 1982). Segregation levels in the Netherlands are also higher relative to those in Chile (Ladd et al., 2009).

²⁰ This system determines whether a student is "priority" based on individual and household surveys collected by the Chilean government. See Elacqua et al. (2009) for details on the decision tree the Ministry of Education uses to classify "priority" students.

important challenge for future research in Chile. For example, while new for-profit schools may choose to either locate in neighborhoods with higher proportions of high school and college educated adults or in poor areas with high concentrations of disadvantaged students, more established Catholic schools are often located in neighborhoods that have become substantially gentrified in recent years (Conferencia Episcopal de Chile, 2002; Sabatini et al., 2010).²³ Neighborhood demographics is a central consideration, since the enrollment practices of schools depend on their ability to access alternative student populations. Given high levels of residential segregation in Chile (Sanhueza and Larranaga, 2008), and absent any policy on the part of the Catholic Church to address student segregation, the student composition of Catholic schools likely reflects demographic trends and stratified residential patterns more than or in addition to socially motivated parental choice and school enrollment practices.

Although it is unclear what the causal factors are that explain segregation patterns, the findings reported in this paper suggest that, in the presence of residential segregation, private school selection, and shared financing, private voucher schools, particularly Catholic schools, respond to market incentives by focusing on a more elite student body, and, in the case of for-profit schools, by stratifying student populations. However, preliminary evidence also suggests that public policies that provide incentives for schools to enroll disadvantaged students can, in some cases, offset some of these effects.

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²³ Gentrification is the process by which higher income households displace lower income residents of a neighborhood (Kristen, 2008).

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